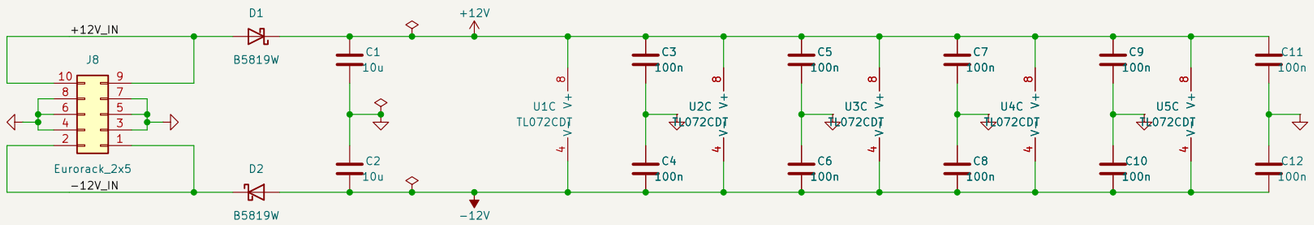
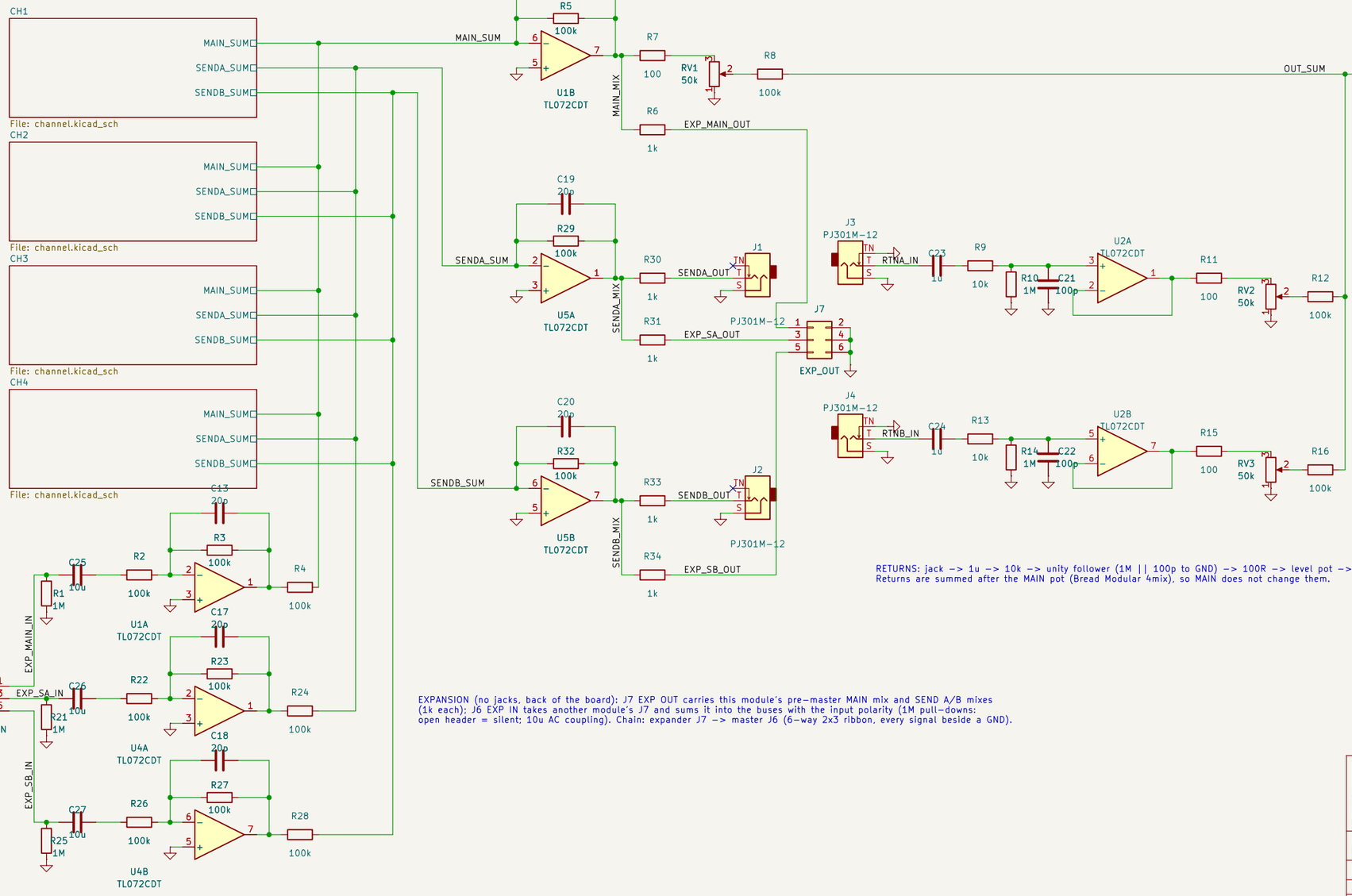
BIG MODULAR 4MIX — S2 schematic: 4-channel mixer with 2 sends / 2 returns, +/-12 V, 18HP (Bread Modular 4mix, no overdrive). Circuit intent: docs/research.md

AUDIO MIXER: every input (channel, return, EXP IN) is AC-coupled; outputs are DC from the op-amps.

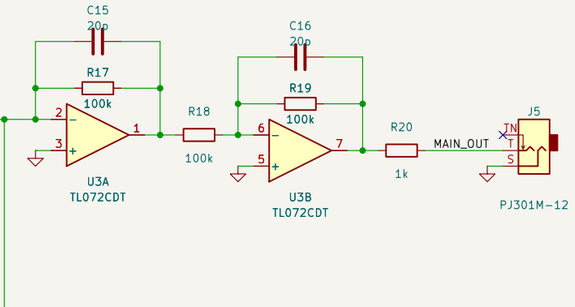
POWER: Eurorack +12V / -12V -> B5819W per rail -> +12V / -12V (about +/-11.6 V). J8 +5V unused.
C3..C12 (root) and Cn02/Cn03 (channels): 100n at every op-amp supply pin; C1/C2: 10u bulk.



CHANNELS: sheet channel.kicad_sch x 4. Each channel ties onto MAIN_SUM / SENDA_SUM / SENDB_SUM
(virtual-ground summing nodes of U1B / U5A / U5B) through its own 100k resistors.



OUTPUT: OUT = MAIN wiper + RETURN A + RETURN B (U3A sum, U3B invert), 1k series.
Unity end to end at full CW; TL072 on +/-11.6 V clips near +/-10 V.



RETURNS: jack -> 1u -> 10k -> unity follower (1M || 100p to GND) -> 100R -> level pot -> 100k -> OUT_SUM.
Returns are summed after the MAIN pot (Bread Modular 4mix), so MAIN does not change them.

EXPANSION (no jacks, back of the board): J7 EXP OUT carries this module's pre-master MAIN mix and SEND A/B mixes
(1k each); J6 EXP IN takes another module's J7 and sums it into the buses with the input polarity (1M pull-downs:
open header = silent; 10u AC coupling). Chain: expander J7 -> master J6 (6-way 2x3 ribbon, every signal beside a GND).

Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S2 — 4-channel audio mixer (AC-coupled inputs), 2 sends / 2 returns, expansion, +/-12

Big Modular

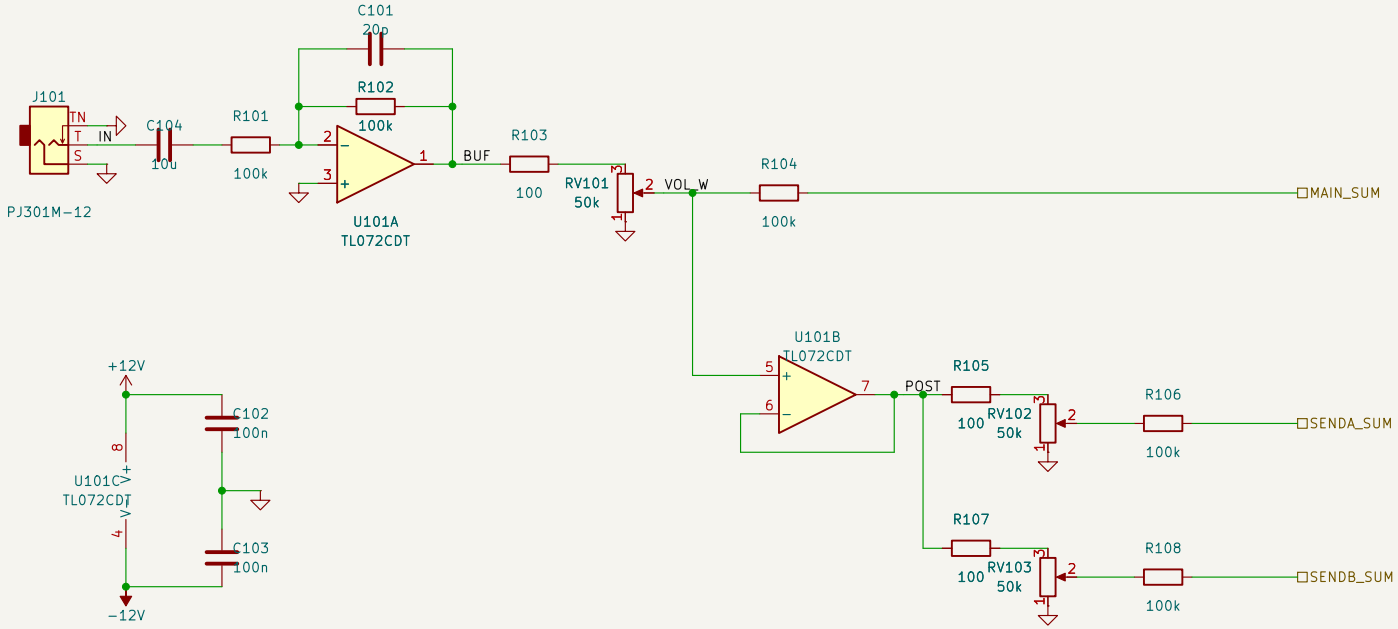
Sheet: /
File: 4mix.kicad_sch

Title: Big Modular — 4MIX

Size: A3	Date:	**Rev: S2**
KiCad E.D.A. 10.0.6		Id: 1/5

CHANNEL STRIP (one sheet, used 4 times: CH1..CH4; parts n01.. per channel)

Bread Modular 4mix channel on +/−12 V: jack −> 10u −> 100k −> inverting buffer (100k || 20p, gain −1) −> 100R −> VOL (CW = pin 3).
VOL wiper −> 100k −> MAIN_SUM. A unity follower buffers the wiper (post−fader) and drives SEND A / SEND B through 100R;
each send wiper −> 100k −> SENDA_SUM / SENDB_SUM. Audio only: AC−coupled (10u into 100k, −3 dB at 0.16 Hz), 100k input
impedance at any knob position; TN grounds an unpatched input. Full CW = unity gain (the root summers invert again, so MAIN/SEND outputs keep the input polarity).



Taper: each wiper sees 100k to a virtual ground (as Bread Modular 4mix/imix), so the 50k B pots run slightly
below linear (about 44 % at mid rotation).

Instantiated 4 times by 4mix.kicad_sch; parts numbered n01.. per channel
SCHEMATIC S2 − one channel: AC−coupled input, input buffer, VOL, post−fader SEND A / SEND B

Big Modular

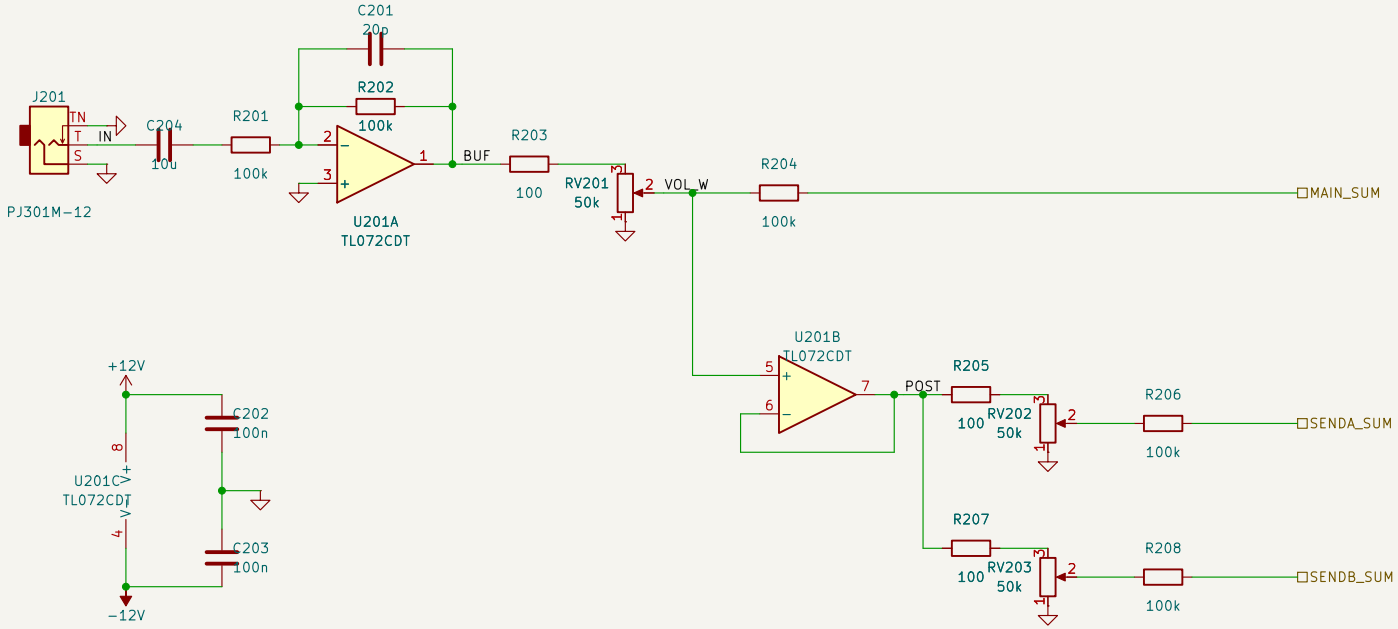
Sheet: /CH1/
File: channel.kicad_sch

Title: Big Modular − 4MIX channel strip (CH1..CH4)

Size: A4	Date:	**Rev: S2**
KiCad E.D.A. 10.0.6		Id: 2/5

CHANNEL STRIP (one sheet, used 4 times: CH1..CH4; parts n01.. per channel)

Bread Modular 4mix channel on +/−12 V: jack −> 10u −> 100k −> inverting buffer (100k || 20p, gain −1) −> 100R −> VOL (CW = pin 3).
VOL wiper −> 100k −> MAIN_SUM. A unity follower buffers the wiper (post−fader) and drives SEND A / SEND B through 100R;
each send wiper −> 100k −> SENDA_SUM / SENDB_SUM. Audio only: AC−coupled (10u into 100k, −3 dB at 0.16 Hz), 100k input
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Taper: each wiper sees 100k to a virtual ground (as Bread Modular 4mix/imix), so the 50k B pots run slightly
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Instantiated 4 times by 4mix.kicad_sch; parts numbered n01.. per channel
SCHEMATIC S2 − one channel: AC−coupled input, input buffer, VOL, post−fader SEND A / SEND B

Big Modular

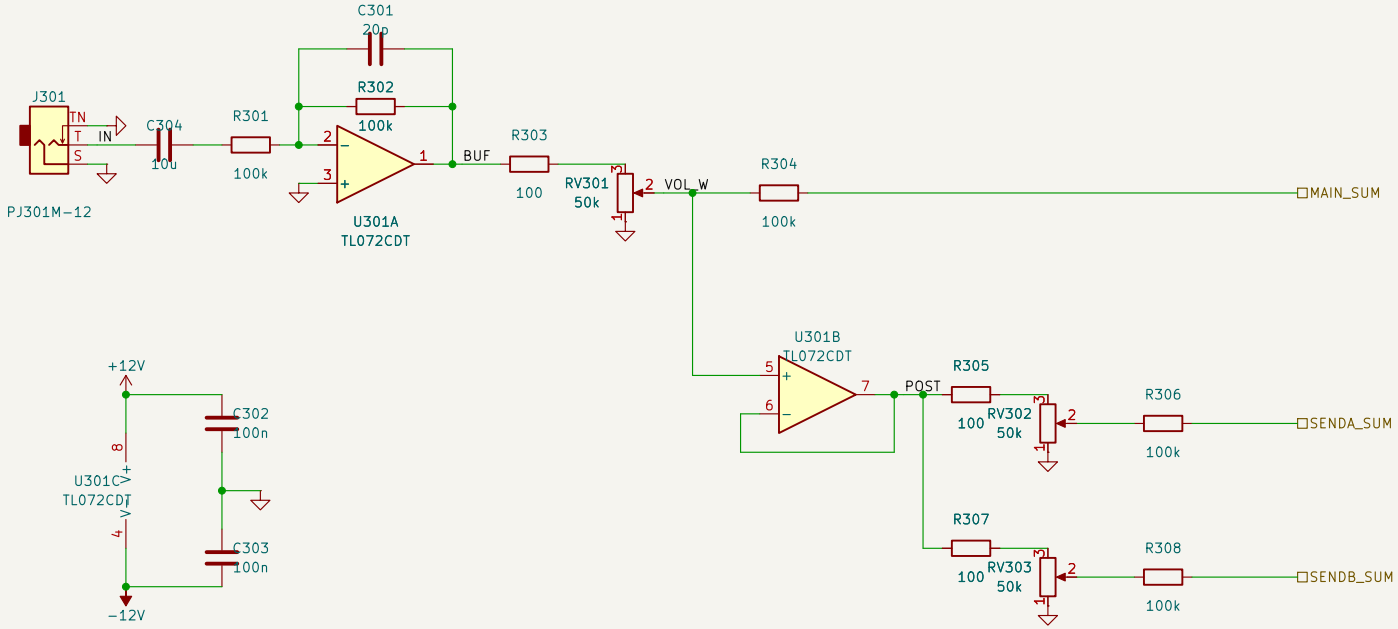
Sheet: /CH2/
File: channel.kicad_sch

Title: Big Modular − 4MIX channel strip (CH1..CH4)

Size: A4	Date:	**Rev: S2**
KiCad E.D.A. 10.0.6		Id: 3/5

CHANNEL STRIP (one sheet, used 4 times: CH1..CH4; parts n01.. per channel)

Bread Modular 4mix channel on +/−12 V: jack −> 10u −> 100k −> inverting buffer (100k || 20p, gain −1) −> 100R −> VOL (CW = pin 3).
VOL wiper −> 100k −> MAIN_SUM. A unity follower buffers the wiper (post−fader) and drives SEND A / SEND B through 100R;
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Taper: each wiper sees 100k to a virtual ground (as Bread Modular 4mix/imix), so the 50k B pots run slightly
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Instantiated 4 times by 4mix.kicad_sch; parts numbered n01.. per channel
SCHEMATIC S2 − one channel: AC−coupled input, input buffer, VOL, post−fader SEND A / SEND B

Big Modular

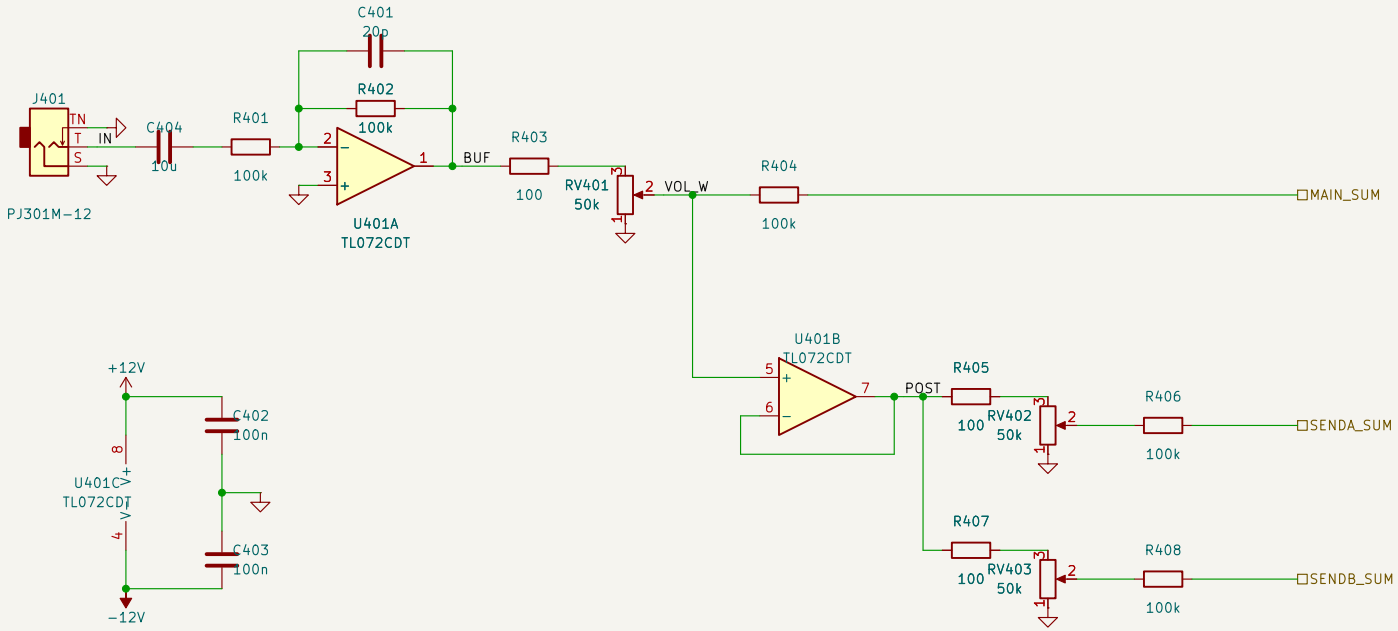
Sheet: /CH3/
File: channel.kicad_sch

Title: Big Modular − 4MIX channel strip (CH1..CH4)

Size: A4	Date:	**Rev: S2**
KiCad E.D.A. 10.0.6		Id: 4/5

CHANNEL STRIP (one sheet, used 4 times: CH1..CH4; parts n01.. per channel)

Bread Modular 4mix channel on +/−12 V: jack −> 10u −> 100k −> inverting buffer (100k || 20p, gain −1) −> 100R −> VOL (CW = pin 3).
VOL wiper −> 100k −> MAIN_SUM. A unity follower buffers the wiper (post−fader) and drives SEND A / SEND B through 100R;
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Taper: each wiper sees 100k to a virtual ground (as Bread Modular 4mix/imix), so the 50k B pots run slightly
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Instantiated 4 times by 4mix.kicad_sch; parts numbered n01.. per channel
SCHEMATIC S2 − one channel: AC−coupled input, input buffer, VOL, post−fader SEND A / SEND B

Big Modular

Sheet: /CH4/
File: channel.kicad_sch

Title: Big Modular − 4MIX channel strip (CH1..CH4)

Size: A4	Date:	**Rev: S2**
KiCad E.D.A. 10.0.6		Id: 5/5